

Floating point: normalisation 1

Advanced: Challenge 1 

Floating point numbers are normalised so that they can be stored with optimum precision.
Which of the following values is **not** properly normalised?

- ☐

mantissa	exponent
1.000000001	0110
- ☐

mantissa	exponent
1.010011110	1111
- ☐

mantissa	exponent
0.011101101	0110
- ☐

mantissa	exponent
0.111010001	0100

Quiz:

[STEM SMART Computer Science Week 33](#)

All teaching materials on this site are available under a CC BY-NC-SA 4.0 license, except where otherwise stated.



Floating point: binary to denary 1

Advanced: Challenge 2  

The binary number shown below is represented as a **normalised floating point number** with an 8-bit mantissa and a 4-bit exponent. The mantissa and exponent are both stored using two's complement.

mantissa										exponent			
0	.	1	1	0	0	1	1	0		0	1	0	0

Convert the floating point number into denary.

Type your answer as a **signed decimal number** (e.g. +3.75) – do not leave any spaces in your answer.

Quiz:
[STEM SMART Computer Science Week 33](#)

All teaching materials on this site are available under a CC BY-NC-SA 4.0 license, except where otherwise stated.



Floating point: binary to denary 3

Advanced: Challenge 2  

The binary number shown below is represented as a **normalised floating point number** with an 8-bit mantissa and a 4-bit exponent. The mantissa and exponent are both stored using two's complement.

mantissa									exponent			
0	.	1	0	1	0	0	0	0	1	0	1	1

Convert the floating point number into denary.

Type your answer as a **signed decimal number**, e.g. +3.75 – do not leave any spaces in your answer.

Quiz:
[STEM SMART Computer Science Week 33](#)

All teaching materials on this site are available under a CC BY-NC-SA 4.0 license, except where otherwise stated.



Floating point: denary to binary 2

Advanced: Challenge 2  

The denary number $-\frac{9}{64}$ (or -0.140625 as a decimal) has been converted to binary, and is held as a **normalised floating point number** with an 8-bit mantissa and a 4-bit exponent. The mantissa and exponent use two's complement.

Which of the following options shows the correct binary representation of the number?

- ☐

mantissa	exponent
1.0111000	0010
- ☐

mantissa	exponent
0.1001000	1110
- ☐

mantissa	exponent
1.1101110	0000
- ☐

mantissa	exponent
1.0111000	1110

Quiz:
STEM SMART Computer Science Week 33

Floating point: denary to binary 1

Advanced: Challenge 2  

Convert the denary number $+\frac{5}{16}$ (or $+0.3125$ as a decimal) to binary, encoding the number as a **normalised floating point number** with an 8-bit mantissa and a 4-bit exponent. The mantissa and exponent use two's complement.

Type your answer as a 12-bit binary number with the binary point (e.g. 0.11100011101) - do not leave any spaces in your answer.

Quiz:

[STEM SMART Computer Science Week 33](#)

All teaching materials on this site are available under a CC BY-NC-SA 4.0 license, except where otherwise stated.



Floating point: number range 1

Advanced: Challenge 2 

A floating point number is represented by a mantissa and an exponent. Both parts are stored using two's complement representation. The floating point number is normalised.

The number of bits allocated to the **mantissa** affects the **precision** of numbers that can be represented, while the **exponent** affects the **range** of the numbers.

If 5 bits are allocated to the mantissa and 3 bits are allocated to the exponent, what is the **largest (most positive) number** that can be represented?

Give your answer as a denary (base-10) number.

Quiz:

[STEM SMART Computer Science Week 33](#)

All teaching materials on this site are available under a CC BY-NC-SA 4.0 license, except where otherwise stated.



Floating point: number range 2

Advanced: Challenge 2 

A floating point number is represented by a mantissa and an exponent. Both parts are stored using two's complement representation. The floating point number is normalised.

The number of bits allocated to the **mantissa** affects the **precision** of numbers that can be represented, while the **exponent** affects the **range** of the numbers.

If 5 bits are allocated to the mantissa and 3 bits are allocated to the exponent, what is the **positive number closest to zero** that can be represented?

Give your answer as a denary (base-10) number.

Quiz:

STEM SMART Computer Science Week 33

All teaching materials on this site are available under a CC BY-NC-SA 4.0 license, except where otherwise stated.



Floating point: number range 4

Advanced: Challenge 2 

A floating point number is represented by a mantissa and an exponent. Both parts are stored using two's complement representation. The floating point number is normalised.

The number of bits allocated to the **mantissa** affects the **precision** of numbers that can be represented, while the **exponent** affects the **range** of the numbers.

If 5 bits are allocated to the mantissa and 3 bits are allocated to the exponent, what is the **negative number closest to zero** that can be represented?

Give your answer as a denary (base-10) number.

Quiz:

[STEM SMART Computer Science Week 33](#)

All teaching materials on this site are available under a CC BY-NC-SA 4.0 license, except where otherwise stated.



Floating point: addition

Advanced: Challenge 2 

Select the correct result for the addition of binary numbers shown below. The binary number representation used is a **normalised floating point number** with an 8-bit mantissa and a 4-bit exponent. The mantissa and exponent are both stored using two's complement.

	mantissa	exponent
	01110110	0011
+	01001011	0100

☐

mantissa	exponent
11000001	0111

☐

mantissa	exponent
010000110	0100

☐

mantissa	exponent
10000110	0100

☐

mantissa	exponent
01000011	0101

Quiz:

[STEM SMART Computer Science Week 33](#)

All teaching materials on this site are available under a CC BY-NC-SA 4.0 license, except where otherwise stated.



Floating point: underflow

Advanced: Practice 2  

A computer system uses a **5-bit** mantissa and **3-bit** exponent for the representation of normalised floating-point numbers. Select which of the results below, arising from a floating point calculation, will cause a floating point **underflow**.

☐ $+\frac{1}{2} \times 2^{-5}$

☐ $+\frac{1}{2} \times 2^{-4}$

☐ -1×2^3

☐ $-\frac{9}{16} \times 2^{-4}$

Quiz:

STEM SMART Computer Science Week 33

All teaching materials on this site are available under a CC BY-NC-SA 4.0 license, except where otherwise stated.

